BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Note: Q.1 Choose the Correct Answers:

Q1. If the temperature of a cold body decreases, efficiency of heat angine wil _____.

(a) Increase
(b) Decrease
(c) Remains same
(d) None of these

Q2. Normal human body temperature on Celsius scale is _____ oC.

(a) 37
(b) 40.6
(c) 98.6
(d) 96.4

Q3. Two parallel conductors carrying current in same direction, they _____.

(a) Attract each other
(b) Repel each other
(c) Neutralize each other
(d) None of these

Q4. Lenz;s Law predicts the _____

(a) Direction of induced emf
(b) Direction of induced current
(c) Magnitude of induced emf
(d) Magnitude of induced current

Q5. S.I Unit of co-efficient of thermal expansion is:

(a) K-1
(b) k
(c) mK
(d) m/k

Q6. By moving a charge of 20 Coulombs 200 J of work is done, then the potential difference between the points.

(a) 10V
(b) 8 V
(c) 5 V
(d) 2 V

Q7. If we increase the distance between tow plates of the capacitor, the capacitance will:

(a) Increase
(b) Decrease
(c) Remains same
(d) First increase then decrease

Q8. The total resistance in a series is equals to:

(a) Product of all resistance
(b) Sum of all resistance
(c) Half of the sum of all resistance
(d) None of these

Q9. Laser is an intense beam of light, which is:

(a) Monochromatic
(b) Coherent
(c) Collimated
(d) All of these

Q10. X-rays have the same essential nature as:

(a) Alpha rays
(b) Beta rays
(c) Gamma rays
(d) Positive rays

Q11. The process of scattering of an X-ray photon by an electron is called:

(a) Compton effect
(b) Photoelectric effect
(c) Pair production
(d) Annihilation

Q12. Half life of a radioactive element can be calculated by formula:

(a) [illegible in source scan]
(b) [illegible in source scan]
(c) [illegible in source scan]
(d) [illegible in source scan]

Q13. Ammeter always connected _____ in circuit.

(a) Perpendicular
(b) Parallel
(c) Series
(d) None of these

Q14. For pair production the minimum energy of photon is _____

(a) 0.76 MeV
(b) 0.511 MeV
(c) 2.11 MeV
(d) 1.022 MeV

Q15. The amount of energy equivalent 1 a.m,u is _____

(a) 931.5 MeV
(b) 0.511 MeV
(c) 2.11MeV
(d) 1.022 MeV

Q16. S.I unit of electric field intensity is:

(a) N/S
(b) N/C
(c) 0.315 MeV
(d) 0.9315 MeV

Q17. The intensity of magnetic field near its pole is _____

(a) Zero
(b) Maximum
(c) Minimum
(d) Variable

Q18. In electromagnetic waves electric and magnetic fields are:

(a) Parallel to each other
(b) Perpendicualr ot each other
(c) Opposite to each other
(d) None of these

Q19. For each material emission occurs only for certain definite frequency of light called:

(a) Critical frequency
(b) Threshold frequency
(c) High frequency
(d) None of these

Q20. Number of electron in one Coulomb charge are:

(a) 9.11 x 10^-31
(b) 1.6 x 10^-10
(c) 8.25 x 10^16
(d) 0.25 x 10^-18

Q21. The nucleus of hydrogen with symbol 1H3 is called:

(a) Proton
(b) Deuterium
(c) Tritum
(d) All of these

Q22. In Isothermal process:

(a) ΔU=ΔW
(b) ΔU = ΔW
(c) VQ = VU
(d) ΔQ = ΔW

Q23. Unstable isotopes are called:

(a) isobars
(b) Isomers
(c) Radioactive
(d) None of these

Q24. Light consists of particles known as:

(a) Protons
(b) Electrons
(c) Neutrons
(d) Photons

Q25. Mass of Positron equal ot mass of:

(a) Proton
(b) Neutron
(c) Electron
(d) Photons

Q26. Unit of Plank’s constant is sane as the unit of:

(a) Energy
(b) Power
(c) Linear momentum
(d) Angular momentum

Q27. S.I unit of electric current is Ampere and one Ampere is equal to:

(a) 1 Joule / sec
(b) 1volt /sec
(c) 1 Coulomb /sec
(d) None of these

Q28. The resistance of a conductor does not depends upon its _____

(a) Length
(b) Area of cross-section
(c) Resistively
(d) Mass

Q29. The smaller unit of magnetic induction is Gauss, and Gauss related to tesla as __

(a) 1T = 10^4G
(b) 1T = 10^-4G
(c) 1T = 10^9G
(d) 1T = 10^3G

Q30. The process in which the original signal recovered is known as:

(a) Demodulation
(b) Detection
(c) Both (a) and (b)
(d) Transmission

Q31. Radio waves of wave length 300 m having frequency of:

(a) 10^-3KHz
(b) 500 KHz
(c) 1MHz
(d) 8 MHz

Q32. A particle with rest mass m0 has total energy 5/3 m0 C^2 the its K.E is ____

(a) 1/3 $m0C^2$
(b) 2/3 $m0C^2$
(c) 4/3 $m0C^2$
(d) 7/3 $m0C^2$

Q33. What will be the resistance, if current flowing through a conductor is 10 amp and voltage is 220 Volts.

(a) 2Ohms
(b) 1.5 Ohms
(c) 0.5 Ohms
(d) 22 Ohms

Q34. A field of 1C is moving perpendicular to the magnetic field of 0.5 tesla with velocity of 10 m/sec.

(a) 5N
(b) 0.5 N
(c) 10 N
(d) 0 N

Q35. Transistor in a circuit basically acts as:

(a) Voltage amplifier
(b) Current amplifier
(c) power amplifier
(d) Rectifier

Q36. If the quantity of charge on each of two bodies is doubled. The force between them becomes:

(a) Twice
(b) Four times
(c) Nine times
(d) Sixteen times

Q37. Kelvion temperature equal to:

(a) 273+ °C
(b) 273 − °C
(c) 273 +°F
(d) 273 - °F

Q38. Two capacitors of capacitance 2C and C respectively are connected in series, their equivalent capacitance becomes.

(a) 2C
(b) C/2
(c) 2C /3
(d) 3C /2

Q39. Electron volt is the unit of:

(a) Work
(b) Frequency
(c) Potential difference
(d) Energy

Q40. Decay constant of a radioactive element is inversely proportional to _____

(a) Activity
(b) Half life
(c) mean life
(d) Wave length

Q41. In P-type semi-conductors the majority of charge carries are:

(a) Electrons
(b) Holes
(c) Neutrons
(d) Protons

Q42. Isobaric changed in a gas show that there is no change in:

(a) Internal energy
(b) Pressure
(c) Volume
(d) Temperature

Q43. Fahrenheit and Celsius scales coincide at:

(a) 273
(b) -40
(c) 0
(d) 32

Section-B (Short Answer)

> Note: Answer any SIX of the following questions.

Q2. A heat engine operators between two reservoirs at temperatures of 25oC and 300°C. What will be its maximum efficiency?

Q3. Why the resistance of conductor is inversely proportional to the area of cross – section of conductor?

Q4. Define electromotive force and electric field intensity. And what are their Units.

Q5. Define hennery and describe the mutual induction of pair o coils with the help o diagram.

Q6. We want to convert galvanometer inot (a) an ammeter and (b) a voltmeter. Wha to do in each cose?

Q7. What is the difference between? Fission and Fusion reaction? Why Fusion reaction is more difficult to achieve?

Q8. Two capacitors of 3μF and 9μF, are joined in series and a potential difference o 200 Volts applied. Find P.D for each capacitor.

Q9. What are the advantages and disadvantages of X-ray? Write some of its applications.

Q10. Explain Why does the resistance of a conductor rise with the rise in temperature?

Section-C (Descriptive Answer)

> Note: Answer any TWO of the following questions.

Q11. Describe the cyclic process of a Carnot Heat engine and derive the expression for its efficiency.

Q12. Determine the equivalent capacitance of three capacitors of capacitance C1, C2 and C3. When connected in (i) series and (ii) Parallel.

Q13. Write notes on any TWO of he following: (a) Photoelectric effect (b) Force on current carrying conductor placed in a uniform magnetic field. (c) Transformer.